



United International University

Department of Computer Science & Engineering

CSE 3715: Data Communication

Class Test 02

Duration: 40 Minutes

Total Marks: 20

ID:

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Convert **last two digits of your student ID** to a **7 bit binary** number:

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Your ID (XY)

Binary Representation of Your ID

1. A data communication channel has a bandwidth of **XY GHz** and a signal-to-noise ratio of 255. If the system operates at 85% of its theoretical maximum capacity, calculate the minimum number of valid signal levels required to support this data rate.

[6]

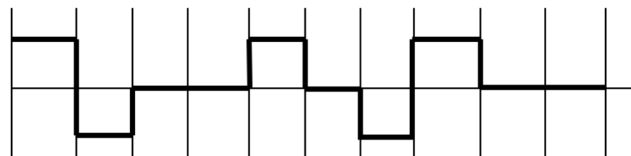
2. A communication cable has an attenuation of 0.5 dB/km. If the signal power at the input of the cable is 5 mW, what will be the signal power after **XY** kilometers?

[6]

3. Convert **binary representation of student ID** into digital signals using Manchester line coding scheme.

[4]

4. Consider the following digital signal encoded using Bipolar AML.



Convert it to a digital signal using **NRZ-I** line coding scheme.
Consider the last signal level was positive.

[4]